Vector Auto-Regression Model for Inflation Forecasting

September 2026

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1 Data Description

Figure 1: Output Gap Timeseries

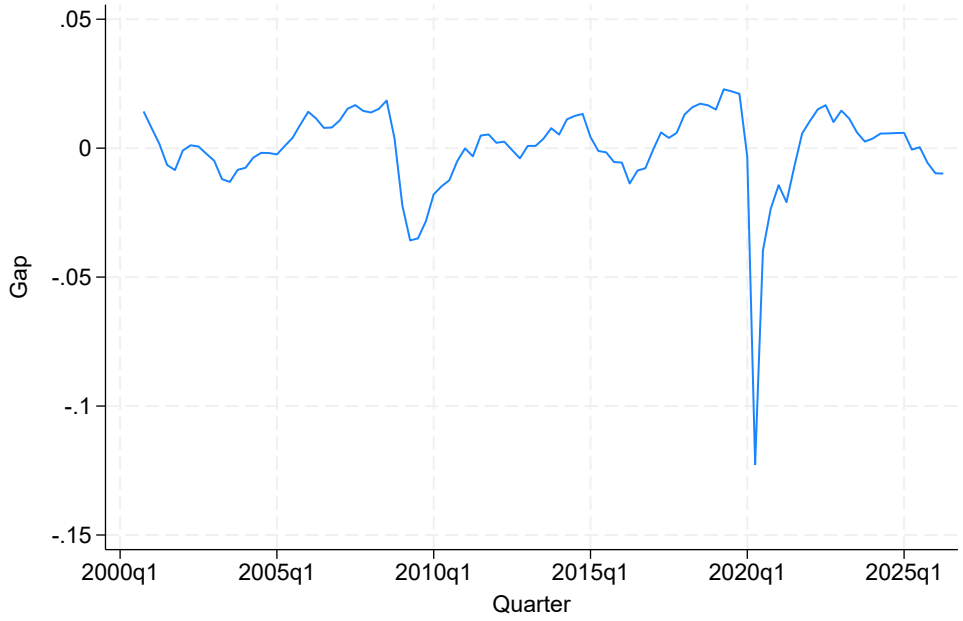


Figure 2: Commodity Price Index Timeseries

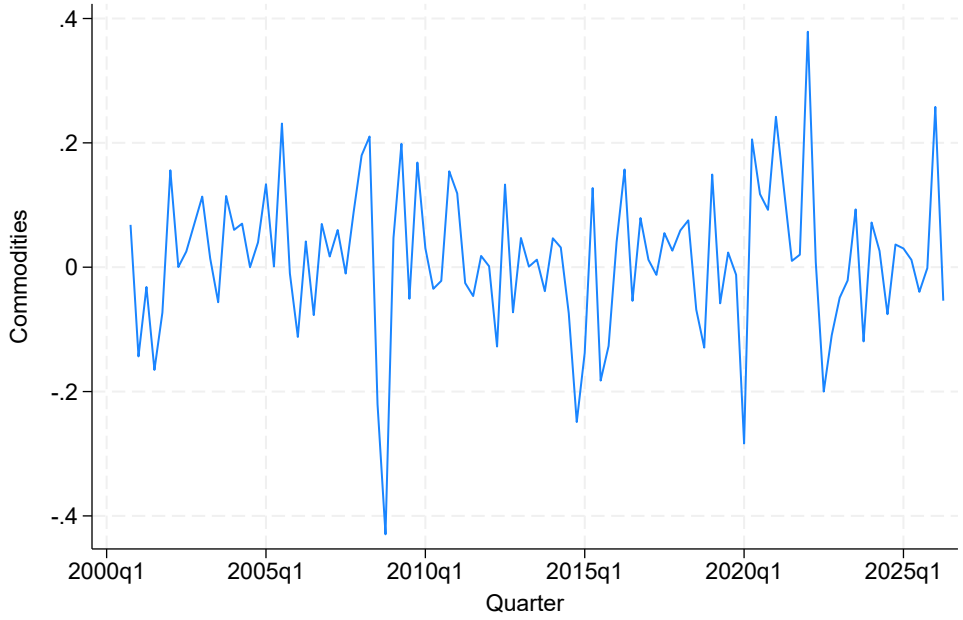


Figure 3: Unemployment Rate

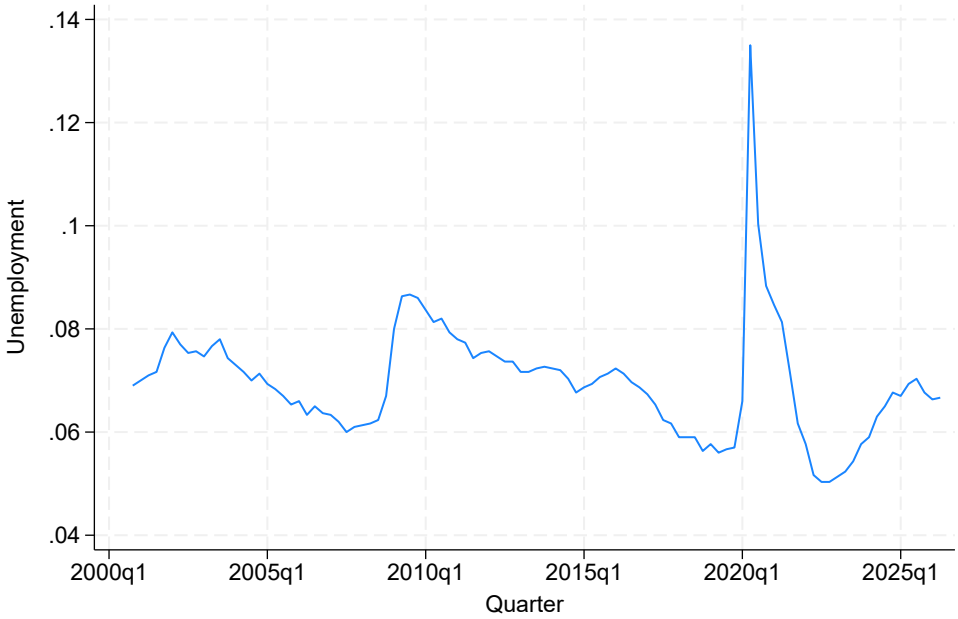


Figure 4: Inflation Rate (Trim)

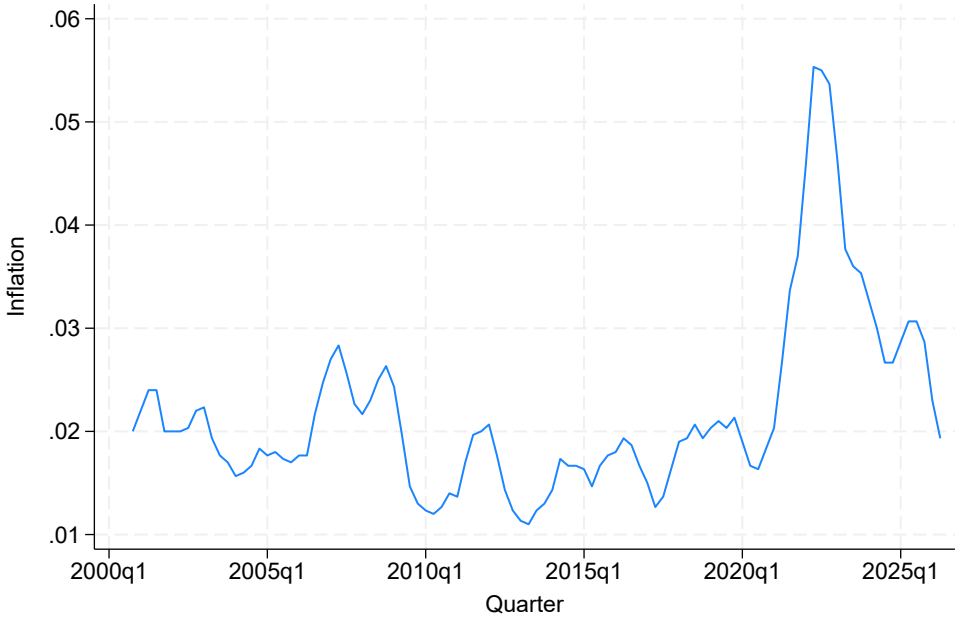


Figure 5: Policy Target Rate

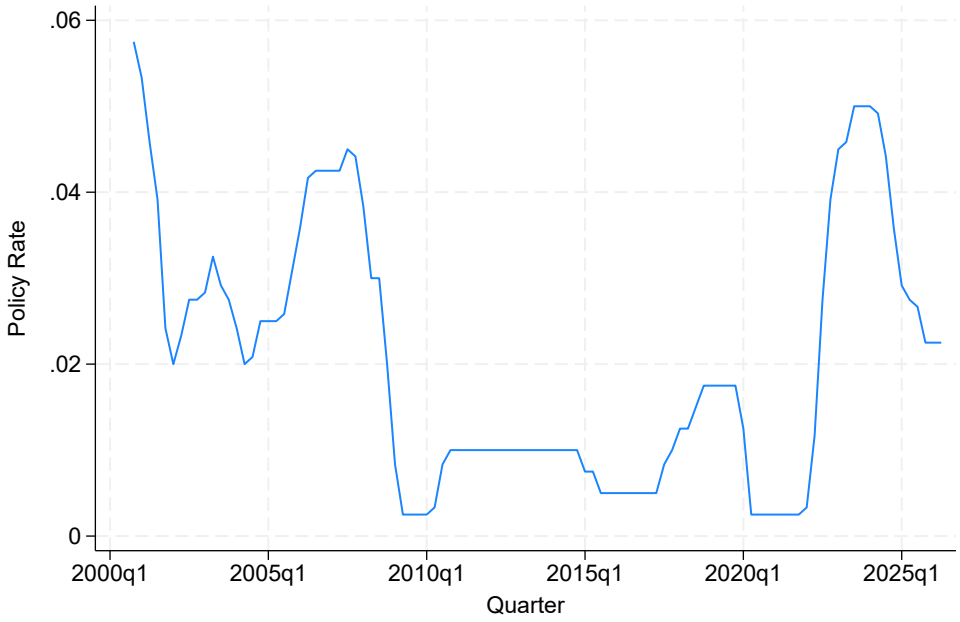


Figure 6: Breakeven Inflation



2 Data Stationarity

Table 1: Output Gap Optimal Lag Test

output_gap_lagsel									
	lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
r1	0	253.7116	.b	.b	.b	.0003196	-5.210548	-5.199815	-5.184004
r2	1	274.3651	41.30699	1	1.30e-10	.0002131	-5.615774	-5.594309	-5.562688
r3	2	274.4369	.1436787	1	.7046509	.0002172	-5.596637	-5.564438	-5.517007
r4	3	274.5034	.1330879	1	.7152517	.0002215	-5.577391	-5.534459	-5.471217
r5	4	274.5382	.0694555	1	.7921312	.0002259	-5.557488	-5.503824	-5.424771
r6	5	275.1973	1.318224	1	.2509111	.0002275	-5.550459	-5.486062	-5.391199
r1	6	276.3844	2.374177	1	.1233566	.0002267	-5.554317	-5.479187	-5.368513

Table 2: Commodity Price Index Optimal Lag Test

com_growth_lagsel									
	lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
r1	0	68.95857	.b	.b	.b	.0144208	-1.401208	-1.390475	-1.374664
r2	1	69.08374	.2503394	1	.6168362	.0146833	-1.38317	-1.361704	-1.330083
r3	2	71.40296	4.638434	1	.0312635	.0142895	-1.41037	-1.378172	-1.33074
r4	3	71.58279	.3596628	1	.5486936	.0145336	-1.39346	-1.350528	-1.287286
r5	4	71.68606	.2065458	1	.6494884	.0148055	-1.37497	-1.321306	-1.242253
r6	5	71.7142	.0562896	1	.8124595	.0151062	-1.354932	-1.290535	-1.195672
r1	6	71.82944	.2304695	1	.6311759	.0153857	-1.336689	-1.261559	-1.150886

Table 3: Unemployment Rate Optimal Lag Test

unemp_rate_lagsel									
	lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
r1	0	296.6848	.b	.b	.b	.0001318	-6.096593	-6.08586	-6.070049
r2	1	332.8658	72.36201	1	1.79e-17	.0000638	-6.821974	-6.800509	-6.768888
r3	2	332.8701	.0086591	1	.9258605	.0000651	-6.801445	-6.769247	-6.721815
r4	3	332.8742	.0083247	1	.9273023	.0000665	-6.780912	-6.737981	-6.674739
r5	4	333.3759	1.003365	1	.3164977	.0000672	-6.770638	-6.716973	-6.637921
r6	5	334.254	1.756179	1	.1851018	.0000673	-6.768124	-6.703727	-6.608864
r1	6	335.0671	1.626157	1	.2022354	.0000676	-6.76427	-6.68914	-6.578466

Table 4: Inflation Rate (Trim) Optimal Lag Test

ifl_trim_lagsel									
	lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
r1	0	317.8277	.b	.b	.b	.0000852	-6.53253	-6.521798	-6.505987
r2	1	434.8797	234.104	1	7.59e-53	7.79e-06	-8.925355	-8.90389	-8.872269
r3	2	463.3918	57.02416	1	4.30e-14	4.42e-06	-9.492615	-9.460416	-9.412985
r4	3	463.9305	1.077438	1	.2992713	4.46e-06	-9.483104	-9.440173	-9.37693
r5	4	463.9527	.0442984	1	.8332995	4.55e-06	-9.462942	-9.409278	-9.330225
r6	5	467.6122	7.318924	1	.0068232	4.31e-06	-9.517776	-9.453379	-9.358516
r1	6	472.8516	10.47888	1	.0012075	3.95e-06	-9.605187	-9.530057	-9.419384

Table 5: Policy Target Rate Optimal Lag Test

pol_rate_lagsel									
	lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
r1	0	273.8904	.b	.b	.b	.0002108	-5.626607	-5.615874	-5.600063
r2	1	402.4158	257.0507	1	7.54e-58	.0000152	-8.255996	-8.23453	-8.202909
r3	2	429.834	54.83646	1	1.31e-13	8.82e-06	-8.800701	-8.768503	-8.721071
r4	3	430.6913	1.714605	1	.190389	8.85e-06	-8.797759	-8.754828	-8.691585
r5	4	432.1948	3.007015	1	.0829048	8.75e-06	-8.808141	-8.754476	-8.675424
r6	5	432.3702	.3506996	1	.5537174	8.91e-06	-8.791138	-8.72674	-8.631877
r1	6	433.4055	2.070617	1	.1501608	8.90e-06	-8.791866	-8.716736	-8.606062

Table 6: Breakeven Inflation Optimal Lag Test

breakeven_lagsel									
	lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
r1	0	391.4553	.b	.b	.b	.0000172	-8.134485	-8.123688	-8.107773
r2	1	486.8891	190.8676	1	2.06e-43	2.40e-06	-10.10186	-10.08026	-10.04843
r3	2	488.7222	3.666243	1	.0555252	2.36e-06	-10.11921	-10.08682	-10.03908
r4	3	490.5682	3.691833	1	.0546795	2.32e-06	-10.13684	-10.09365	-10.02999
r5	4	490.5887	.0411822	1	.8391866	2.37e-06	-10.11643	-10.06245	-9.982872
r6	5	491.2291	1.280678	1	.257773	2.38e-06	-10.10894	-10.04415	-9.948668
r1	6	491.2793	.1003778	1	.7513768	2.43e-06	-10.08915	-10.01357	-9.902168

Table 7: Dickey-Fuller Test Table

	dfuller_results				
	lags	N	Zt	Zt_trend	pvalue
output_gap	1	101	-4.691359	-4.669291	.0000874
com_growth	0	102	-9.446612	-9.401005	4.74e-16
unemp_rate	1	101	-3.704348	-3.847716	.0040524
ifl_trim	6	96	-2.49713	-2.90804	.1161741
pol_rate	2	100	-2.887516	-2.832724	.046819
breakeven	1	100	-2.440323	-3.069722	.1306773

3 Forecast

Table 8: VAR Model Optimal Lag Test

	model_lagsel								
	lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
r1	0	-916.8248	.b	.b	.b	9.009308	19.22552	19.2903	19.38579
r2	1	-455.9149	921.8198	36	3.8e-170	.0012908	10.37323	10.82672	11.49513
r3	2	-388.1428	135.5442	36	1.85e-13	.0006716	9.711308	10.55351	11.79484
r4	3	-357.688	60.90949	36	.0058688	.0007703	9.826834	11.05774	12.872
r5	4	-332.83	49.71611	36	.0637751	.0010132	10.05896	11.67857	14.06575
r6	5	-309.64	46.37995	36	.1152296	.0014192	10.32583	12.33415	15.29426
r1	6	-273.9857	71.30866	36	.0004112	.0015926	10.33304	12.73006	16.26309

Figure 7: VAR Model Unit Root Test

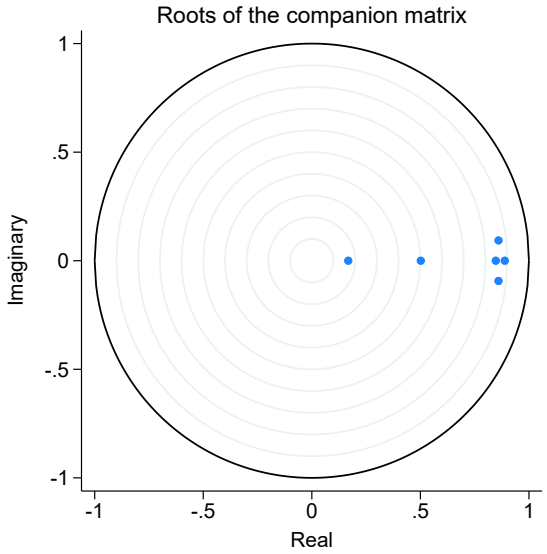


Figure 8: VAR Model Inflation Forecast (8 quarters)

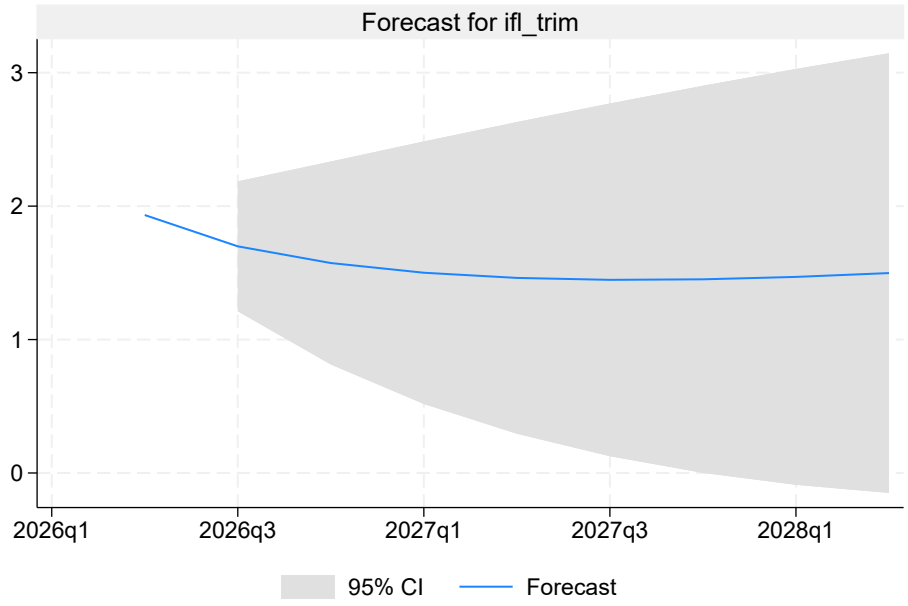


Figure 9: Impulse Response Function of Inflation to a Policy Hike

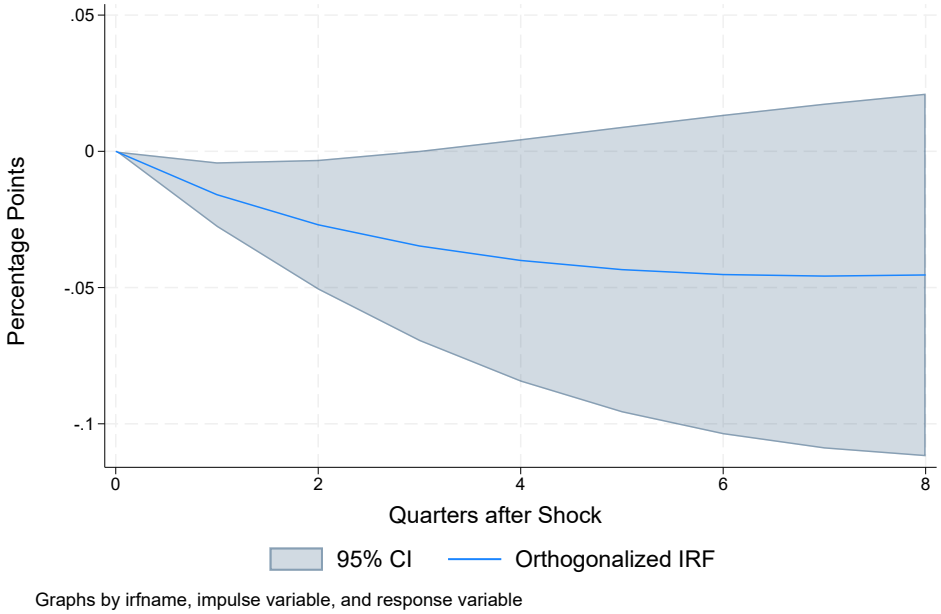
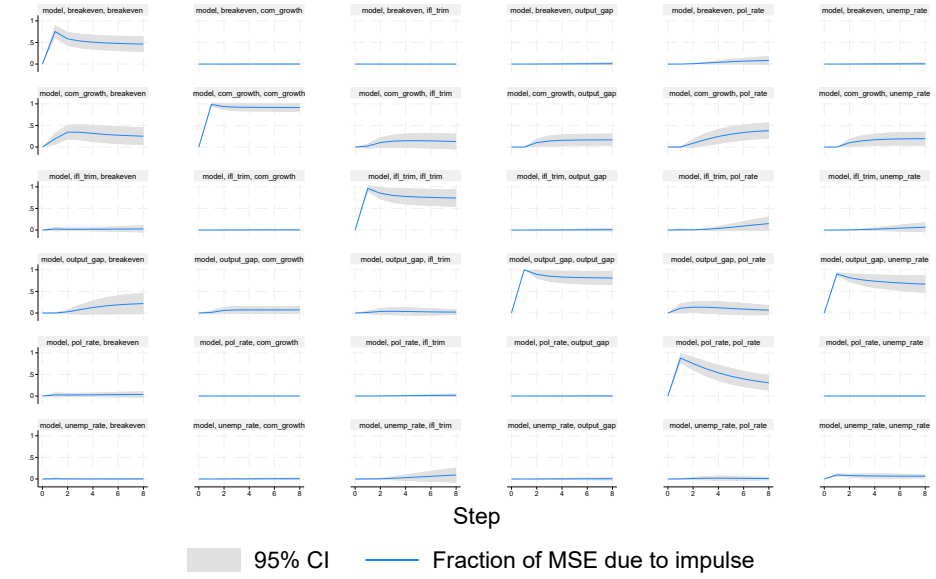


Figure 10: Forecast Error Variance of Model



Graphs by irfname, impulse variable, and response variable